Certified Commitment Horizons in Dynamic Lot Sizing

Prefix Stability, Inverse Shortest Paths and Calibrated Forecast Revisions

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Verification status. The mathematical claims have a producer-side proof and code audit. The release includes deterministic tests, raw synthetic results and one-command replay. It has not received external peer review, independent clean-room reproduction or proof-assistant verification. The unrestricted complexity classification of WW-Prefix-Radius$_\infty$ remains open.

Abstract

A rolling lot-sizing plan becomes operationally costly when every forecast revision changes orders that staff or suppliers have begun to execute. Classical planning-horizon work asks how much future information is needed before early decisions stabilise. We ask a different inverse question: given the current forecast and a selected optimal Wagner–Whitin schedule, how large must the forecast revision become before any optimal schedule can leave its first k replenishment cycles?

We formalise this quantity as a strong prefix-stability radius in a weighted L_∞ norm, with coordinate-wise demand floors, and lead with the associated decision problem WW-Prefix-Radius$_\infty$. The Wagner–Whitin condition reduces every arc to $K_i + q_i(D_j - D_i)$ with ordered slopes. This yields a Monge inequality, an exact fixed-path tie formula, and a polynomial frozen-comparator lower certificate computed by $O(T)$ linear-fractional shortest-path programmes. The lower certificate is exact for a full schedule but can be strict for a partial prefix; a three-period positive-demand example has certified radius 1.5 and exact radius 2.

For exact analysis, each outside path induces a compact inverse linear programme whose suffix dual represents the best prefix-retaining continuation. Their finite disjunction gives the exact radius, while a fixed-radius mixed-integer formulation decides WW-Prefix-Radius$_\infty$ with $O(T^3)$ product variables. We prove membership in NP but leave the unrestricted exact complexity open. Historical forecast revisions can calibrate the deterministic radius through a finite-sample split-conformal threshold. Finally, Bellman residuals telescope across rolling replans, giving a deterministic regret bound with a certified dead zone: only revision magnitude beyond the stability radius is charged. Synthetic checks reproduce the polynomial certificate in 569 exhaustive cases and find no rolling-bound violation beyond 2.28×10^{-13} in 220 trials. These computations audit implementation; they do not establish representativeness or substitute for the proofs.

Keywords: dynamic lot sizing; Wagner–Whitin; forecast revisions; nervousness; stability radius; inverse optimisation; shortest paths; conformal calibration; rolling horizon.

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1 The decision problem

Suppose a planner has solved a deterministic single-item lot-sizing problem under the current demand forecast $\widehat{d} \in \mathbb{R}_+^T$. The chosen optimum is a sequence of setup epochs, equivalently a shortest path. Some first replenishment cycles are about to be released to a supplier, warehouse or production team. The planner needs a quantitative answer to one question:

> How far must the demand forecast move before a rationally optimal schedule can change any protected setup epoch or cycle boundary?

A fixed seven- or fourteen-day fence does not answer this question. It ignores the current cost geometry, the size and direction of plausible revisions, and the fact that the best continuation after a protected prefix can itself change. We therefore define a distance to the nearest revised instance at which an optimal path can leave the prefix.

Periods are $1, \dots, T$. A path $P = (v_0 = 1 < v_1 < \cdots < v_m = T + 1)$ has an arc (v_r, v_{r+1}) for each replenishment cycle. We select one nominal optimal path

$$P^\star = (v_0 = 1 < v_1 < \cdots < v_m = T + 1)$$

and protect its first k arcs, denoted $\pi_k = (v_0, \dots, v_k)$. Let

$$\mathcal{I}_k = \{P : P \text{ starts with } \pi_k\}, \qquad \mathcal{O}_k = \mathcal{P} \setminus \mathcal{I}_k.$$

For revised demand d, define the best inside and outside values

$$A_k(d) = \min_{P \in \mathcal{I}_k} C_P(d), \qquad B_k(d) = \min_{Q \in \mathcal{O}_k} C_Q(d).$$

The protected prefix remains a property of every optimal schedule exactly when $B_k(d) > A_k(d)$. Equality already permits a rational schedule change, so deployment rules must treat the boundary strictly.

Let $w \in \mathbb{R}_{++}^T$ and

$$\|x\|_{\infty,w} = \max_{1 \le t \le T} w_t |x_t|.$$

The weights can encode units, forecast-horizon relevance or asymmetric operational scaling after an appropriate coordinate transformation. Let $\ell \in \mathbb{R}_+^T$ be a demand floor satisfying $\ell \le \widehat{d}$ coordinate-wise.

Definition 1.1 (Strong prefix-stability radius)**.** The weighted radius of the first k replenishment cycles is

$$\rho_{k,w}(\widehat{d}) = \inf \left\{ \left\| d - \widehat{d} \right\|_{\infty,w} : d \ge \ell,\ B_k(d) \le A_k(d) \right\}. \tag{1}$$

For $w = \mathbf{1}$ we write $\rho_{k,\infty}$.

The definition is deliberately strong. For every $r < \rho_{k,w}$, every optimiser under every admissible revision of weighted size at most r retains the prefix. The radius concerns setup epochs and cycle boundaries. It does not freeze literal order quantities: quantities change when demand changes even if the setup pattern does not.

Decision problem 1.2 (WW-Prefix-Radius$_\infty$)**. Input:** a rational Wagner–Whitin instance, rational $\widehat{d}, \ell, w, R$, a selected nominal optimal path $P^\star$, and $k \in \{1, \dots, m\}$.
Question: does there exist $d \ge \ell$ such that

$$\left\| d - \widehat{d} \right\|_{\infty,w} \le R \quad \text{and} \quad B_k(d) \le A_k(d)?$$

Equivalently, is $\rho_{k,w}(\widehat{d}) \le R$? The unweighted problem sets $w = \mathbf{1}$.

This decision problem leads the paper because it states the operational object directly. The later lower certificate, exact disjunction, statistical calibration and rolling-loss theorem each answer a distinct part of it.

Proposition 1.3 (Elementary geometry)**.** *Assume $P^\star$ is nominally optimal.*

(a) If $r < \rho_{k,w}(\widehat{d})$, every optimal path at every admissible d with $\left\|d - \widehat{d}\right\|_{\infty,w} \le r$ belongs to $\mathcal{I}_k$.

(b) If $\rho_{k,w}(\widehat{d}) < \infty$, the nearest destabilising point is attained. Unless $\rho_{k,w} = 0$, at least one nearest point satisfies $A_k(d) = B_k(d)$.

(c) $\rho_{k+1,w}(\widehat{d}) \le \rho_{k,w}(\widehat{d})$ whenever both prefixes exist.

(d) A nominal outside tie implies $\rho_{k,w}(\widehat{d}) = 0$.

Proof. Part (a) restates the complement of the event in (1). For (b), restrict attention to a closed weighted ball whose radius exceeds the infimum; its intersection with $\{d : d \ge \ell\}$ is compact. The functions A_k and B_k are minima of finitely many affine path costs and hence continuous, so a convergent destabilising sequence attains the infimum. If the nearest point had $B_k < A_k$, continuity along its line segment from $\widehat{d}$ would produce an earlier equality. For (c), $\mathcal{I}_{k+1} \subseteq \mathcal{I}_k$ and $\mathcal{O}_{k+1} \supseteq \mathcal{O}_k$, so $A_{k+1} \ge A_k$ and $B_{k+1} \le B_k$; the destabilising set can only expand. Part (d) follows by evaluating the event at $\widehat{d}$. □

2 Position relative to earlier horizon and sensitivity work

Wagner and Whitin (1958) formulate deterministic single-item dynamic lot sizing and show how its structure supports a forward algorithm and planning-horizon arguments. Subsequent algorithms exploit concavity, geometric envelopes and Monge structure to obtain $O(T \log T)$ performance in general variants and linear time under Wagner–Whitin costs (Federgruen and Tzur 1991; A. Wagelmans et al. 1992; Aggarwal and Park 1993).

The planning- and forecast-horizon literature asks whether extending the terminal horizon or changing unspecified future data can alter early decisions. Protective procedures and rolling-horizon error bounds address the cost of finite look-ahead (Eppen et al. 1969; Lundin and Morton 1975; Lee and Denardo 1986). Bes and Sethi (1988) distinguish conditional forecast and decision horizons. Federgruen and Tzur (1994) and Federgruen and Tzur (1995) compute minimal forecast horizons and, where needed, adapt full-horizon plans to control worst-case optimality gaps under desired stability conditions. Integer-programming and discrete-demand approaches extend this line (Dawande et al. 2006; Dawande et al. 2007).

More recent stochastic work models forecast evolution inside rolling-horizon lot sizing and evaluates the value of anticipating later forecast updates (Forel and Grunow 2023). That line changes the forward planning model. Our inverse object instead conditions on a realised current forecast and certifies how far the next revision must move before a specified deterministic prefix can cease to be compulsory.

Our object differs in three respects. First, it conditions on the current finite forecast rather than an unknown extension of the horizon. Second, it measures a simultaneous norm-bounded revision to all forecast coordinates. Third, it protects a class of paths that share a prefix while allowing the best continuation inside that class to switch.

The closest deterministic antecedent is the post-optimality analysis of Hoesel and A. P. M. Wagelmans (1993), which computes parameter ranges that retain an optimal economic lot-sizing solution. Generic stability-radius work studies the size of objective perturbation needed to change a combinatorial optimum (Chakravarti and A. P. M. Wagelmans 1998); inverse optimisation asks for the smallest cost perturbation that makes a specified solution optimal (Ahuja and Orlin 2001). Here demand revisions

induce correlated changes across many arcs, and the protected object is a prefix face rather than one fixed path.

A search frozen on 4 August 2026 covered publisher records, DOI searches, accessible manuscripts and English title/abstract queries for these literatures. It did not locate a source combining the prefix radius, a floor-aware polynomial lower certificate, exact inverse and mixed-integer models, finite-sample forecast-revision calibration, and a rolling loss bound with a stability dead zone. This is a bounded negative search result, not proof of absolute priority. The full search boundary and strings appear in the release.

3 Reduced cumulative-demand arcs and ordered slopes

3.1 From physical costs to a shortest path

Let $K_i \geq 0$ be the setup cost in period i, p_i the unit production or purchase cost, and $h_i \geq 0$ the cost of carrying one unit from period i to $i+1$. Demand $d_t \geq 0$ must be met without backlogging. Under the no-speculative-motive condition

$$p_i + h_i \geq p_{i+1} \qquad (i = 1, \ldots, T-1), \tag{2}$$

an optimal plan has the zero-inventory-ordering structure used by the Wagner–Whitin shortest path.

Define

$$H_i = \sum_{s=1}^{i-1} h_s, \qquad q_i = p_i - H_i, \qquad D_1 = 0, \qquad D_j = \sum_{t=1}^{j-1} d_t.$$

Condition (2) is equivalent to

$$q_1 \geq q_2 \geq \cdots \geq q_T. \tag{3}$$

An arc (i, j) means that a setup in period i covers periods $i, \ldots, j-1$.

Proposition 3.1 (Reduced cumulative-demand arc form)**.** *The physical cost of arc (i, j) is*

$$K_i + \sum_{t=i}^{j-1} \left(p_i + \sum_{s=i}^{t-1} h_s \right) d_t.$$

For every complete path this equals the path-independent term $\sum_{t=1}^{T} H_t d_t$ plus reduced arc costs

$$\bar{c}_{ij}(d) = K_i + q_i(D_j - D_i) = K_i + q_i \sum_{t=i}^{j-1} d_t. \tag{4}$$

Consequently reduced and physical costs have the same optimal paths and the same path-to-path regret.

Proof. For $t \geq i$,

$$p_i + \sum_{s=i}^{t-1} h_s = p_i + H_t - H_i = q_i + H_t.$$

Summing over the periods assigned to all path arcs contributes $\sum_t H_t d_t$ exactly once, independently of the path, and leaves (4). □

The canonical graph starts at vertex 1. If early demand can be zero, a path may encode a vacuous setup. A literal operational interpretation should either impose a strictly positive floor, preprocess zero-demand periods, or state an explicit convention. This paper retains a general non-negative floor in the mathematics and uses a positive floor in the strict counterexample.

3.2 Monge structure and what it does not solve

Proposition 3.2 (Ordered slopes imply Monge arcs)**.** *For $i < i' < j < j'$,*

$$\bar{c}_{ij}(d) + \bar{c}_{i'j'}(d) \le \bar{c}_{ij'}(d) + \bar{c}_{i'j}(d). \tag{5}$$

Proof. After cancellation, the left side minus the right side is

$$(q_{i'} - q_i)(D_{j'} - D_j) \le 0$$

by (3) and non-negative demand. □

This ordered-slope representation is the first proof attack on WW-Prefix-Radius$_\infty$. It yields affine path coefficients, the Monge inequality, additive perturbation geometry and an exact coefficient diameter used later. It does not by itself make the exact prefix-radius problem a shortest path. The exact event compares two lower envelopes, $B_k(d)$ and $A_k(d)$, and the path attaining $A_k(d)$ can switch before an outside path wins. A natural continuous path-flow relaxation of the exact mixed-integer model is not integral: the three-period instance in section 6 has true radius 2, while the relaxation accepts radius 1. This negative result rules out treating the obvious relaxation as a certificate; it does not prove hardness.

3.3 Affine path costs

For a path P, let $i_P(t)$ be the setup period whose cycle contains t. Then

$$C_P(d) = \kappa_P + \beta_P^\top d, \qquad \kappa_P = \sum_{i \in P \setminus \{T+1\}} K_i, \qquad \beta_{P,t} = q_{i_P(t)}. \tag{6}$$

This affine form converts forecast stability into distance from $\widehat{d}$ to unions of polyhedra.

Proposition 3.3 (Complexity upper bound)**.** *The rational decision problem WW-Prefix-Radius$_\infty$ belongs to NP.*

Proof. A yes-certificate specifies an outside path Q, a revised rational vector d in the radius box, and shortest-path dual potentials for the suffix that follows the protected prefix. The potentials satisfy polynomially many inequalities of the form in (20) below, and the candidate inequality verifies $C_Q(d) \le A_k(d)$. For fixed Q, these conditions form a rational linear programme of polynomial size. Whenever it is feasible, it has a basic feasible solution with polynomially bounded encoding length. A verifier checks the path, demand bounds, weighted box, dual inequalities and candidate inequality in polynomial time. □

No matching hardness result or polynomial exact algorithm is claimed. The exact complexity of the unrestricted problem is an explicit open question in this release.

4 Exact distance to a fixed competitor

Fix an outside path Q. Relative to $P^\star$, define its nominal gap and coefficient difference

$$g_Q = C_Q(\widehat{d}) - C_{P^\star}(\widehat{d}) \ge 0, \qquad b_Q = \beta_Q - \beta_{P^\star}. \tag{7}$$

Let $s_t = \widehat{d}_t - \ell_t$ be the available downward slack. For any coefficient vector b, define

$$\Psi_b(r) = r \sum_{t:b_t<0} \frac{|b_t|}{w_t} + \sum_{t:b_t>0} b_t \min\left\{\frac{r}{w_t}, s_t\right\}. \tag{8}$$

The first term increases coordinates with negative b_t; the second decreases coordinates with positive b_t until the floor binds.

Theorem 4.1 (Fixed-path tie radius)**.** *The exact distance from $\widehat{d}$ to the half-space $C_Q(d) \le C_{P^\star}(d)$ under $d \ge \ell$ is*

$$r(Q, P^\star) = \inf\{r \ge 0 : \Psi_{b_Q}(r) \ge g_Q\}. \tag{9}$$

At radius r, a worst revision is

$$d_t(r) = \begin{cases} \max\{\ell_t, \widehat{d}_t - r/w_t\}, & b_{Q,t} > 0, \\ \widehat{d}_t + r/w_t, & b_{Q,t} < 0, \\ \widehat{d}_t, & b_{Q,t} = 0. \end{cases} \tag{10}$$

Before any floor binds,

$$r(Q, P^\star) = \frac{g_Q}{\sum_t |b_{Q,t}|/w_t}, \tag{11}$$

provided the denominator is positive and the resulting revision remains above the floor.

Proof. For a fixed radius, minimising the affine gap

$$g_Q + b_Q^\top (d - \widehat{d})$$

over the weighted box and floor constraints separates by coordinate. A positive coefficient is reduced by decreasing demand as far as the box and floor permit; a negative coefficient is reduced by increasing demand to the upper side of the box. The maximum attainable gap reduction is exactly $\Psi_{b_Q}(r)$. The half-space is reached exactly when that reduction is at least g_Q. Formula (11) follows while all positive-coefficient coordinates remain unsaturated. □

The function Ψ_b is continuous, non-decreasing and piecewise linear. Its breakpoints are $w_t(\widehat{d}_t - \ell_t)$ for coordinates with $b_t > 0$. Thus a fixed path needs only sorting and a linear scan; the challenge is selecting the relevant outside path without enumeration.

5 A polynomial lower certificate

5.1 Frozen-comparator radius

Define

$$\underline{\rho}_{k,w}(\widehat{d}) = \min_{Q \in O_k} r(Q, P^\star). \tag{12}$$

This freezes the inside comparator at the selected nominal path rather than allowing the best prefix-retaining continuation to switch.

Theorem 5.1 (Validity and endpoint exactness)**.** *For every protected prefix,*

$$\underline{\rho}_{k,w}(\widehat{d}) \le \rho_{k,w}(\widehat{d}). \tag{13}$$

The sequence $\underline{\rho}_{k,w}$ is non-increasing in k. If $k = m$, so the protected prefix is the full path $P^\star$, then

$$\underline{\rho}_{m,w} = \rho_{m,w}.$$

Proof. At any destabilising d, choose $Q \in \mathcal{O}_k$ attaining $B_k(d)$. Then

$$C_Q(d) = B_k(d) \le A_k(d) \le C_{P^\star}(d).$$

Therefore d lies in at least one pairwise half-space used in (12); its distance cannot be below the minimum pairwise distance. Monotonicity follows because $\mathcal{O}_k$ expands with k. When the full path is protected, $\mathcal{I}_m = \{P^\star\}$ and $A_m = C_{P^\star}$, so the pairwise union is the exact destabilising set. □

The lower certificate is operationally useful: whenever a calibrated revision radius lies below it, the true prefix is safe. Its conservatism costs commitment length, not validity.

5.2 Prefix-deviation graph

We now compute (12) without enumerating 2^{T-1} paths. Construct an expanded acyclic graph $\mathcal{G}_k$. Its matching layer contains states that have followed the first $r < k$ nominal arcs. Taking a non-expected arc moves permanently to a deviated layer. The expected kth arc is omitted because taking it completes the protected prefix. The deviated layer contains the ordinary complete acyclic lot-sizing graph. Source-to-sink paths in $\mathcal{G}_k$ are therefore in one-to-one correspondence with $\mathcal{O}_k$. The graph has $O(T)$ vertices and $O(T^2)$ arcs.

Let X_k be its unit source-to-sink flow polytope, $Nx = h$, $x \ge 0$. Because N is a network matrix, its extreme points are path incidence vectors.

For each period define the floor breakpoint

$$\delta_t = w_t(\widehat{d}_t - \ell_t). \tag{14}$$

Sort the distinct positive breakpoints and append 0 and $+\infty$. On an interval $[\delta_\ell, \delta_{\ell+1}]$, the set of positive coefficient differences already saturated at the floor is fixed.

For an expanded arc $e = (i, j)$ and $t \in \{i, \dots, j-1\}$, put $c_{e,t} = q_i - \beta_{P^\star,t}$. Define additive arc intercepts and slopes

$$a_e^\ell = K_i + \sum_{t=i}^{j-1} \begin{cases} c_{e,t}\widehat{d}_t, & c_{e,t} < 0, \\ c_{e,t}\widehat{d}_t, & c_{e,t} > 0 \text{ and } \delta_t > \delta_\ell, \\ c_{e,t}\ell_t, & c_{e,t} > 0 \text{ and } \delta_t \le \delta_\ell, \\ 0, & c_{e,t} = 0, \end{cases} \tag{15}$$

$$b_e^\ell = \sum_{t=i}^{j-1} \begin{cases} -c_{e,t}/w_t, & c_{e,t} < 0, \\ c_{e,t}/w_t, & c_{e,t} > 0 \text{ and } \delta_t > \delta_\ell, \\ 0, & \text{otherwise.} \end{cases} \tag{16}$$

For path vector x, the worst frozen-comparator gap throughout this interval is

$$F_x(r) = (a^\ell)^\top x - \kappa_{P^\star} - r(b^\ell)^\top x. \tag{17}$$

Theorem 5.2 (Polynomial frozen-comparator certificate)**.** *The value $\underline{\rho}_{k,w}$ is computable by at most $T + 1$ polynomial-size linear programmes and shortest-path calculations on $\mathcal{G}_k$.*

On interval $[\delta_\ell, \delta_{\ell+1}]$, first solve a shortest path for $F_x(\delta_\ell)$. If its minimum is positive, the first crossing in that interval is the linear-fractional path problem

$$\min_{x \in X_k :\, (b^\ell)^\top x > 0} \frac{(a^\ell)^\top x - \kappa_{P^\star}}{(b^\ell)^\top x}. \tag{18}$$

It has the Charnes–Cooper linear-programming form

$$\begin{aligned} \min_{y,\tau} \quad & (a^\ell)^\top y - \kappa_{P^\star}\tau \\ s.t. \quad & Ny = \tau h, \\ & (b^\ell)^\top y = 1, \\ & y \geq 0, \quad \tau \geq 0. \end{aligned} \tag{19}$$

The first interval whose optimum lies within its endpoints contains the exact lower certificate.

Proof. By theorem 4.1, the pairwise gap for every outside path is affine in r between consecutive floor breakpoints. Equations (15)–(17) distribute its intercept and slope over the path arcs. Before the first crossing, every path has $F_x(\delta_\ell) > 0$, so any path with positive slope crosses at the ratio in (18); paths with zero slope cannot cross in the interval. Minimising that ratio over the integral path polytope gives the earliest outside path. The substitution $y = \tau x$, $\tau = 1/(b^\ell)^\top x$ yields (19), following Charnes and Cooper (1962). There are at most T finite breakpoints, and each graph and linear programme has polynomial size. □

Remark 5.3 (Numerical versus formal certification)**.** *The theorem is an exact rational linear-programming statement. The release implementation calls floating-point HiGHS through SciPy (Huangfu and Hall 2018; Virtanen et al. 2020). A safety-critical deployment should reconstruct rational primal-dual certificates or round the returned lower radius outward. The supplied number is a tested numerical lower certificate, not a proof object from a formally verified solver.*

6 The lower certificate can be strict

The moving inside continuation is not a technical nuisance; it changes the answer.

Proposition 6.1 (Positive-floor strict-gap instance)**.** *Let $T = 3$, unit weights and*

$$q = (2, 1, 0), \qquad K = (0, 3, 2), \qquad \widehat{d} = (1, 5, 1), \qquad \ell = (0.5, 0.5, 0.5).$$

The selected nominal optimum is $P^\star = (1, 2, 4)$ with reduced cost 11. Protect the first arc $(1, 2)$. Then

$$\underline{\rho}_{1,\infty} = 1.5 \qquad \textit{but} \qquad \rho_{1,\infty} = 2.$$

Proof. The four path costs are

$$\begin{aligned} P^\star = (1,2,4): &\quad 2d_1 + d_2 + d_3 + 3, \\ D = (1,2,3,4): &\quad 2d_1 + d_2 + 5, \\ C = (1,3,4): &\quad 2d_1 + 2d_2 + 2, \\ A = (1,4): &\quad 2d_1 + 2d_2 + 2d_3. \end{aligned}$$

Paths $P^\star, D$ are inside; C, A are outside. The nominal gap between C and $P^\star$ is 3 and their coefficient difference is $(0, 1, -1)$, so theorem 4.1 gives radius $3/2$. At the frozen tie $(1, 3.5, 2.5)$, however, $D = 10.5 < C = P^\star = 11$. The prefix remains optimal.

Now take any admissible d with $\left\|d - \widehat{d}\right\|_\infty < 2$. Then $d_2 > 3$. If $d_3 \geq 2$, D is the best inside path and $C - D = d_2 - 3 > 0$. If $1 \leq d_3 < 2$, $P^\star$ is best inside, C is best outside, and $C - P^\star = d_2 - d_3 - 1 > 0$. If $d_3 < 1$, $P^\star$ is best inside, A is best outside, and $A - P^\star = d_2 + d_3 - 3 > 0$ because $d_3 \geq 0.5$. Thus no radius below 2 destabilises the prefix. At $d = (1, 3, 3)$, whose distance is 2, $D = C = 10$ while $P^\star = 11$ and $A = 14$. Hence the exact radius is 2. □

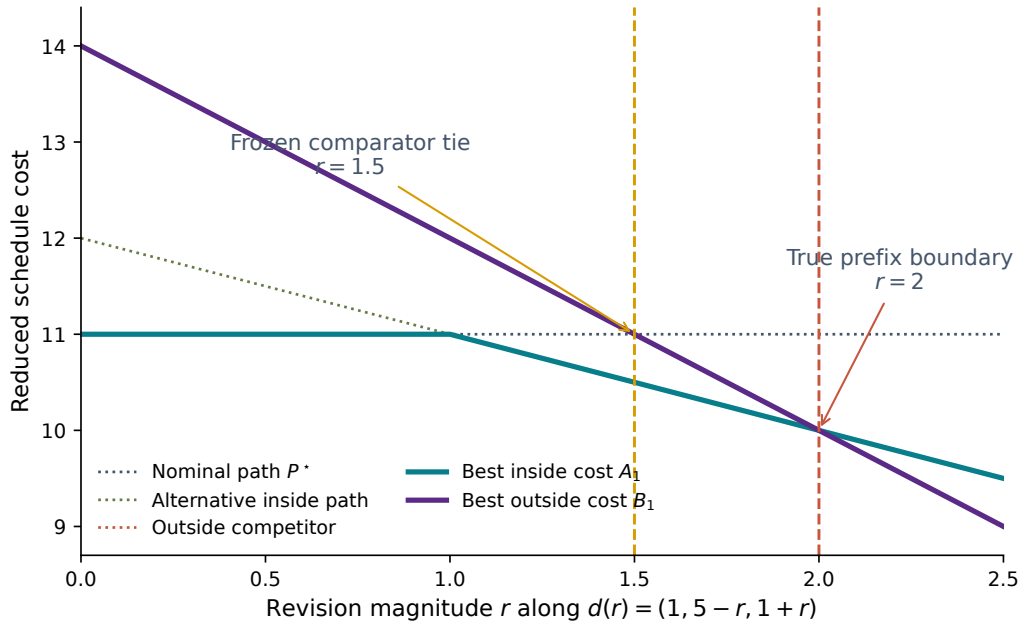


Figure 1: Along one adversarial revision ray, the outside competitor ties the frozen nominal path at radius 1.5, but an alternative inside continuation has already become cheaper. The strong prefix boundary occurs at radius 2. The proof of theorem 6.1 covers the full L_∞ ball, not only this ray.

The example also defeats the continuous flow relaxation of the exact mixed-integer model: at radius 1 it admits a half-integral path-flow solution even though the true event is impossible below radius 2. This shows why the polynomial method is presented as a lower certificate rather than as an exact relaxation.

7 Exact inverse formulations

7.1 A compact linear programme for one outside path

Fix $Q \in \mathcal{O}_k$ and let $s = v_k$ be the last vertex of the protected prefix. Write the prefix cost as

$$C_{\pi_k}(d) = \kappa_{\pi_k} + \beta_{\pi_k}^\top d.$$

The best inside value is this fixed prefix plus a shortest suffix path from s to $T + 1$. Introduce suffix potentials $\pi_s, \dots, \pi_{T+1}$ satisfying

$$\pi_s = 0, \qquad \pi_j - \pi_i \le K_i + q_i \sum_{t=i}^{j-1} d_t \quad (s \le i < j \le T + 1). \tag{20}$$

The maximum feasible π_{T+1} equals the suffix shortest-path value.

Theorem 7.1 (Fixed-competitor inverse LP)**.** *For a fixed outside path Q, its exact inverse radius relative*

to the prefix class is the optimum of

$$\begin{aligned} \min_{d,r,\pi} \quad & r && (21)\\ \text{s.t.} \quad & d_t \geq \ell_t && (t = 1,\dots,T),\\ & -r/w_t \leq d_t - \widehat{d}_t \leq r/w_t && (t = 1,\dots,T),\\ & \pi \text{ satisfies (20)},\\ & C_Q(d) \leq C_{\pi_k}(d) + \pi_{T+1},\\ & r \geq 0. \end{aligned}$$

Denote this value r_Q^{inv}. *Then*

$$\rho_{k,w}(\widehat{d}) = \min_{Q \in O_k} r_Q^{\mathrm{inv}}. \tag{22}$$

Proof. For fixed d, feasible potentials have π_{T+1} no larger than the suffix shortest-path value, and the shortest-path dual has an optimal feasible potential attaining equality. Therefore the last inequality is feasible exactly when $C_Q(d) \leq A_k(d)$. Minimising the radius gives the distance to that fixed disjunct. Taking the minimum over all outside paths gives the union in theorem 1.1. □

Equation (22) is an exact finite disjunctive model. It also supports a certifying enumerator. Sort outside paths by their pairwise lower bounds $r(Q, P^\star)$. Solve (21) in that order, maintain the smallest upper bound U, and stop when the next pairwise lower bound is at least U. Theorem 5.1 makes the pruning valid. The worst case remains exponential, but the recorded small-instance median processed only 6.25% of outside paths.

7.2 An exact fixed-radius mixed-integer model

At a proposed radius R, define the coordinate box

$$L_t = \max\{\ell_t, \widehat{d}_t - R/w_t\}, \qquad U_t = \widehat{d}_t + R/w_t. \tag{23}$$

Put binary unit-flow variables z_e on the prefix-deviation graph $\mathcal{G}_k$. For each arc $e = (i, j)$ and covered period $t = i, \dots, j-1$, introduce $u_{e,t} = d_t z_e$. The binary McCormick hull over $L_t \leq d_t \leq U_t$ is

$$L_t z_e \leq u_{e,t} \leq U_t z_e, \tag{24}$$

$$d_t - U_t(1 - z_e) \leq u_{e,t} \leq d_t - L_t(1 - z_e). \tag{25}$$

The selected outside-path cost is

$$\sum_{e=(i,j)} K_i z_e + \sum_{e=(i,j)} q_i \sum_{t=i}^{j-1} u_{e,t}. \tag{26}$$

Use (20) for the inside suffix. One may impose candidate cost no greater than prefix plus π_{T+1}, or equivalently minimise that cost gap and test whether the optimum is non-positive.

Theorem 7.2 (Fixed-radius MILP)**.** *The flow constraints on* $\mathcal{G}_k$, *bounds* (23), *product constraints* (24)–(25), *suffix potentials* (20), *and the comparison of* (26) *with* $C_{\pi_k}(d) + \pi_{T+1}$ *form an exact mixed-integer formulation of WW-*Prefix-Radius$_\infty$ *at radius* R. *It has* $O(T^3)$ *variables and constraints in the direct construction.*

Proof. Every integer source-to-sink flow on $\mathcal{G}_k$ selects exactly one outside path, and every outside path has one such flow. With binary z_e, (24)–(25) are the exact convex hull of $u_{e,t} = d_t z_e$ over the fixed box, so (26) is its true path cost. By shortest-path duality, the potential comparison holds exactly when the selected outside path is no more expensive than the best prefix-retaining path. The number of graph arcs is $O(T^2)$; summing an auxiliary product over every period covered by every arc gives $O(T^3)$ products. □

The implementation minimises the cost gap rather than relying on a zero-slack feasibility row, then independently re-solves the selected disjunct as (21). This rejects false positives caused by mixed-integer primal-feasibility tolerances near the exact boundary. A feasible witness is directly checkable. A negative solver result still requires retention of status, logs and, where available, an infeasibility certificate.

8 Calibrating forecast revisions

The deterministic radius answers how far demand must move. A commitment policy also needs a probability model for how far the next forecast revision will move. Rather than impose a parametric multivariate error distribution, score historical aligned revisions by the same geometry used in the optimisation.

Let $E_1, \dots, E_n \in \mathbb{R}^T$ be historical revision vectors for the same product class, lead-time grid and forecast vintage. Define

$$S_i = \|E_i\|_{\infty,w}\,. \tag{27}$$

For target miscoverage $\alpha \in (0,1)$, let

$$j = \lceil (n+1)(1-\alpha) \rceil\,, \qquad q_{1-\alpha} = \begin{cases} S_{(j)}, & j \le n, \\ +\infty, & j = n+1, \end{cases} \tag{28}$$

where $S_{(j)}$ is the jth order statistic of the calibration scores.

Theorem 8.1 (Finite-sample revision tube)**.** *If $S_1, \dots, S_n, S_{n+1}$ are exchangeable, then*

$$\Pr\{S_{n+1} \le q_{1-\alpha}\} \ge 1-\alpha. \tag{29}$$

Proof. Under exchangeability, the rank of the next score among the $n+1$ scores is uniform when ties are randomly broken and is conservative under the non-randomised weak inequality. The probability that the rank exceeds j is at most $(n+1-j)/(n+1) \le \alpha$. □

Corollary 8.2 (Calibrated commitment)**.** *If*

$$q_{1-\alpha} < \underline{\rho}_{k,w}(\widehat{d}). \tag{30}$$

then, under the assumptions of theorem 8.1,

$$\Pr\{\textit{the next revised optimum can leave the first } k \textit{ cycles}\} \le \alpha.$$

A conservative policy commits the largest k satisfying (30).

Proof. The event $S_{n+1} \le q_{1-\alpha}$ implies a revision strictly inside the deterministic lower certificate and hence inside the true radius by theorem 5.1. Its complement has probability at most α. □

The strict inequality matters because a tie at the radius permits a schedule change. The weights should be fixed before calibration or fitted on a separate sample. Revision rows must align in units, horizon, vintage and information set. Missing coordinates require an explicit treatment before scoring; silently dropping them changes the norm and invalidates the claimed tube.

Exchangeability is an assumption, not a property that conformal prediction manufactures. The synthetic scale-shift experiment in section 10 deliberately violates it: empirical coverage falls from 0.896 at equal scale to 0.639 and 0.370 when the next revision scale is multiplied by 1.25 and 1.5. Weighted or rolling conformal methods may reduce sensitivity to drift, but they require their own assumptions and error accounting (Barber et al. 2023).

9 A rolling-horizon loss theorem

A stability radius controls decision changes. We now bound the cost of acting on successive forecasts.

At a state a, let $V_a(d)$ be the optimal reduced suffix cost from a through T. Suppose the planner implements a forecast-chosen block $\pi : a \to b$ and replans at b. Its Bellman residual under realised demand is

$$R_\pi(d) = C_\pi(d) + V_b(d) - V_a(d) \geq 0. \tag{31}$$

If implemented blocks concatenate to a path R, their residuals telescope.

Lemma 9.1 (Telescoping identity)**.** *For block endpoints* $1 = a_0 < a_1 < \cdots < a_M = T + 1$*,*

$$C_R(d) - V_1(d) = \sum_{r=0}^{M-1} R_{\pi_r}(d). \tag{32}$$

Proof. Sum (31). The value terms cancel, $V_{T+1} = 0$, and the block costs sum to the implemented path cost. □

For suffix paths P, Q from a, define the weighted coefficient diameter

$$\Lambda_a = \max_{P,Q:a\to T+1} \sum_{t=a}^{T} \frac{|\beta_{P,t} - \beta_{Q,t}|}{w_t}. \tag{33}$$

Lemma 9.2 (Exact ordered-slope diameter)**.** *Under* (3)*,*

$$\Lambda_a = \Gamma_a := \sum_{t=a}^{T} \frac{q_a - q_t}{w_t}. \tag{34}$$

The direct suffix path and the path that sets up every period attain the maximum.

Proof. At period t, any suffix path uses a setup epoch $i \in \{a, \ldots, t\}$, so ordered slopes imply $q_a \geq \beta_{P,t} = q_i \geq q_t$. Hence every pairwise coefficient difference is at most $q_a - q_t$ coordinate-wise. The direct path has coefficient q_a in every period, while the every-period path has coefficient q_t at period t, attaining the sum. □

Theorem 9.3 (Certified dead-zone rolling loss)**.** *At state a, let forecast f choose a prefix block $\pi : a \to b$. Suppose its strong weighted prefix radius is ρ over the admissible demand domain. Then for every realised d,*

$$R_\pi(d) \leq \Lambda_a \left(\|d - f\|_{\infty,w,[a:T]} - \rho \right)_+ . \tag{35}$$

If only a valid lower certificate $\underline{\rho} \le \rho$ is available, replacing ρ by $\underline{\rho}$ preserves the inequality. Consequently, for a rolling policy using forecasts f^r and certified blocks starting at a_r,

$$C_R(d) - V_1(d) \le \sum_r \Gamma_{a_r} \left(\|d - f^r\|_{\infty,w,[a_r:T]} - \underline{\rho}_r \right)_+ . \tag{36}$$

Proof. The residual is the difference between the best path constrained to retain π and the unrestricted optimum. For two demand vectors d, e, choose an inside optimiser at e and an unrestricted optimiser at d to obtain

$$R_\pi(d) - R_\pi(e) \le (\beta_P - \beta_Q)^\top (d - e) \le \Lambda_a \|d - e\|_{\infty,w} .$$

Swapping d, e gives absolute Lipschitz continuity. Throughout the closed radius ball around f, an inside optimum remains globally optimal, so $R_\pi = 0$. If d lies outside, project it radially to the ball at

$$e = f + \frac{\rho}{\|d - f\|_{\infty,w}} (d - f).$$

The segment remains in the floor-constrained domain, and $\|d - e\|_{\infty,w} = \|d - f\|_{\infty,w} - \rho$. Lipschitz continuity proves (35). A smaller lower certificate enlarges the positive part. Sum the block bounds using theorem 9.1 and substitute theorem 9.2. □

Without stability information, the generic Lipschitz bound charges $\Gamma_a \|d - f\|$ from zero. Theorem 9.3 removes a certified dead zone. It is a worst-case bound and may remain conservative; its value is that every term is auditable and it becomes exactly zero when realised revisions stay within each certified radius.

A related approximate-policy statement is immediate. If a chosen block has forecast residual at most ε, then

$$R_\pi(d) \le \varepsilon + \Lambda_a \|d - f\|_{\infty,w} .$$

The stronger dead-zone form requires exact prefix optimality and a valid radius.

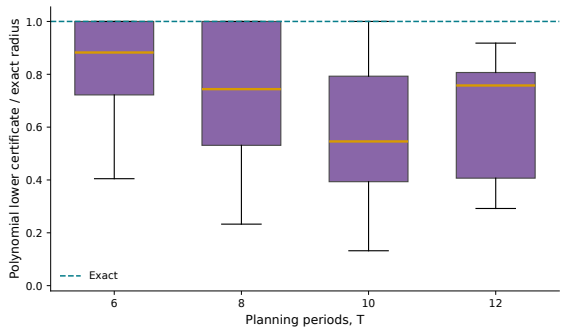
10 Computational audit

The computations target implementation errors, not theorem discovery by simulation. All pseudo-random experiments use NumPy's `default_rng` with seed 20260804. The frozen environment was Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, pandas 2.2.3 and Matplotlib 3.10.8 on x86-64 Linux. Raw comma-separated tables, scripts and figures accompany the paper.

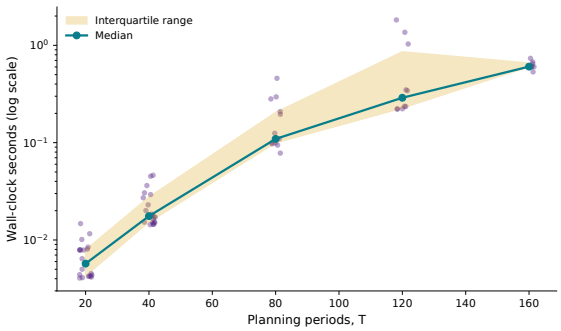
Synthetic Wagner–Whitin instances use strictly decreasing slopes generated from positive increments, setup costs sampled independently, and positive demand. These distributions are test generators, not a model of industrial demand.

The exact MILP and disjunctive LP were cross-checked on random instances with $T = 3, \dots, 6$. A separate explicit LP containing one inequality for every inside path matched the compact suffix-dual LP. At each tested exact radius, the mixed-integer model was infeasible below a tolerance-separated boundary and returned a directly checked witness at the boundary. The implementation now validates the selected mixed-integer path with the independent fixed-competitor LP to avoid a solver-tolerance false positive observed 10^{-5} below one random boundary during adversarial testing.

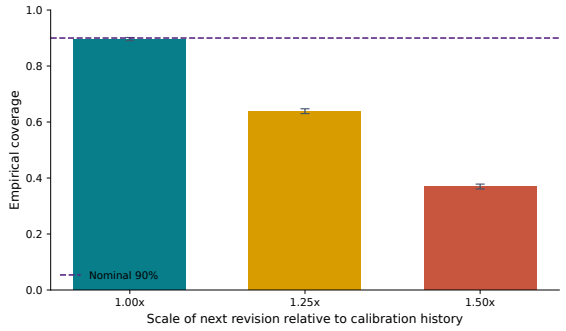
The certificate-tightness distribution makes the intended use clear. The lower bound is often informative but not uniformly close. An organisation should therefore use it as a cheap first-line commitment certificate and invoke exact analysis only when the business value of an additional cycle justifies the computation.



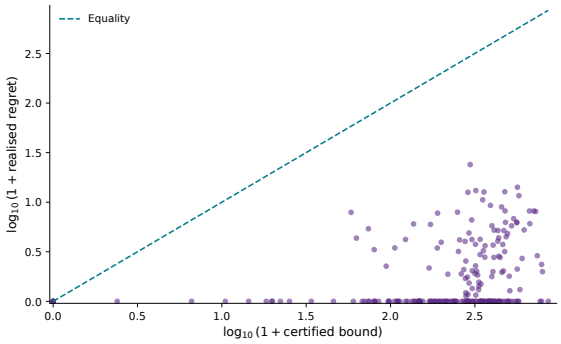
(a) Lower-certificate tightness.



(b) Polynomial runtime.



(c) Calibration and scale drift.



(d) Realised regret versus bound.

Figure 2: Synthetic audit results. Panel (a) compares the polynomial lower certificate with exact strong radii, not with exhaustive computation of the same lower certificate. Panel (b) uses a logarithmic time axis. Panel (c) illustrates the exchangeability boundary. Panel (d) is plotted on $\log_{10}(1 + x)$ axes.

Table 1: Frozen computational audit. Timing is machine-dependent; algebraic comparisons and seeded samples are replayable under the locked environment.

Check	Recorded result	Interpretation
Polynomial vs exhaustive lower certificate	569 cases; maximum absolute error 0	Checks path automaton, floor breakpoints and fractional LP against path enumeration for $T = 2, \ldots, 9$.
Exact small-instance radii	129 finite instances	First-cycle certificate exactly tight in 27.9%; median lower/exact ratio 0.778; tenth percentile 0.396.
Pruned exact enumeration	Median 6.25% of outside paths solved	Observed pruning only; no polynomial worst-case claim.
Polynomial scaling	Median 0.0057 s at $T = 20$; 0.604 s at $T = 160$	Single-machine synthetic timing; not a complexity proof.
Conformal equal-scale coverage	0.8964, Monte Carlo SE 0.00278	Consistent with nominal 0.90 within simulation error.
Conformal scale shift	0.6387 at 1.25×; 0.3697 at 1.5×	Deliberate exchangeability failure.
Rolling theorem	220 cases; maximum excess 2.28×10^{-13}	Numerical round-off only; median certified bound 298.2 versus 719.2 without the dead zone.

11 Operational protocol

The complete system separates deterministic geometry, statistical frequency and realised severity.

Step 1. Freeze the instance. Record $K, q, \widehat{d}$, the demand floor, norm weights, units, forecast vintage and data hash. Verify ordered slopes or document why the reduced Wagner–Whitin model is inapplicable.

Step 2. Solve the nominal problem. Select and record one tie-broken optimal path $P^\star$. Enumerate its replenishment cycles; do not describe changing quantities as changing structural commitments.

Step 3. Compute polynomial certificates. For each k, calculate $\underline{\rho}_{k,w}$. Check monotonicity and retain the primal path, adversarial revision and solver record.

Step 4. Calibrate revision magnitude. Assemble aligned historical revision vectors, audit missingness and drift, compute $q_{1-\alpha}$, and commit the largest k with $q_{1-\alpha} < \underline{\rho}_{k,w}$.

Step 5. Escalate near the boundary. If a longer commitment has material value but fails the lower-certificate test, run the exact fixed-radius MILP or the pruned disjunctive LP. Treat timeouts and uncertified infeasibility as unresolved, not as safe.

Step 6. Replan and monitor. After each forecast revision, rerun the calculation from the current state. Accumulate the deterministic rolling bound in (36) and compare it with realised regret when actual demand becomes available.

Step 7. Govern exceptions. Record manual overrides, capacity constraints, minimum order quantities, supplier lock-ins and service constraints outside the model. A mathematical certificate for the wrong feasible set is operationally invalid.

Table 2: Four outputs that should not be conflated.

Output	Meaning	What it does not mean
$\underline{\rho}_{k,w}$	Deterministic safe lower radius	It need not equal the exact radius.
Exact witness or $\rho_{k,w}$	Nearest revised instance at which an outside optimum can tie or win	It is not a probability of revision.
$q_{1-\alpha}$	Finite-sample marginal revision threshold under exchangeability	It is not a conditional guarantee under arbitrary drift.
Rolling bound	Worst-case realised cost above the clairvoyant optimum	It is not a forecast of expected financial loss without a stochastic model.

The command-line release implements two main operations. `wwprefix certify` returns nominal paths, lower certificates, optional exact small-instance radii and an optional conformal commitment length. `wwprefix decide` solves WW-Prefix-Radius$_\infty$ at one proposed radius. JSON inputs and outputs make each decision auditable and suitable for embedding in a wider planning system.

12 Scope, failure modes and open problems

The conclusions are bounded by the following assumptions.

Model boundary. The theory covers one uncapacitated item, deterministic period demands once a scenario is fixed, linear production and holding costs, no backlogging, and Wagner–Whitin ordered slopes. Capacity, joint setups, minimum order quantities, perishability, substitution and service constraints change the path family and may destroy the formulas.

Demand-domain boundary. The revision set has a coordinate-wise lower floor and no upper cap beyond the radius box. A physical setup interpretation needs positive demand or preprocessing of zero-demand periods. Additional upper bounds can be inserted into the pairwise and exact programmes but require revised breakpoint formulas.

Decision boundary. The prefix fixes structural setup epochs and cycle boundaries. It does not freeze order quantities, labour allocations, transport bookings or supplier-specific decisions unless those are explicitly represented.

Statistical boundary. The split-conformal statement is marginal and requires exchangeable scores. Product pooling, changing forecast systems, promotions, censored demand and missing coordinates can break alignment. The scale-shift experiment is a warning, not a drift correction.

Computational boundary. The polynomial lower certificate is numerically solved in floating point. Exact disjunctive enumeration can be exponential. Mixed-integer infeasibility depends on solver assurance. The release contains no independent reproduction or formal proof.

Open exact complexity. The central unresolved theoretical question is whether the ordered-slope, prefix-class structure makes unrestricted WW-Prefix-Radius$_\infty$ polynomially solvable, or whether the moving inside lower envelope encodes an NP-hard disjunction. Productive next attacks include: (i) a stronger Monge-compatible extended formulation; (ii) a reduction that preserves ordered q and positive demand; (iii) fixed-parameter algorithms in the number of protected cycles, distinct slopes or floor breakpoints; and (iv) approximation guarantees for $\underline{\rho}_{k,w}/\rho_{k,w}$ under cost separation assumptions.

Calibration and control extensions. A second research line should derive time-uniform or drift-robust revision tubes and integrate them with sequential commitment decisions. The deterministic theorem is modular: any valid revision region contained in a stability ball can replace split conformal calibration. A third line should test whether the rolling dead-zone bound predicts operational nervousness and expediting cost on forecast-vintage data from real supply chains.

13 Conclusion

The relevant commitment horizon is not a fixed number of calendar periods. It is the distance from the current forecast to the nearest demand revision at which the best schedule can leave a specified replenishment prefix.

The reduced arc form $K_i + q_i(D_j - D_i)$ makes that distance tractable in layers. Ordered slopes yield Monge structure and a sharp coefficient diameter. Fixed-path geometry yields a closed-form floor-aware tie radius. A prefix-deviation graph turns the minimum of those distances into a polynomial lower certificate. Exact fixed-competitor and mixed-integer formulations resolve cases where that certificate is too conservative. Historical revisions convert the radius into a calibrated commitment rule under an explicit exchangeability assumption. Bellman residuals then convert the same radius into a rolling cost bound with a certified dead zone.

This layered result is operationally complete even while the unrestricted exact complexity remains open. It offers a cheap safe certificate, an exact escalation route, a statistically calibrated decision threshold and a deterministic account of realised rolling loss. Each layer exposes its own assumptions and failure modes, so a planner can distinguish what has been proved, what has been estimated and what remains unresolved.

A Full proof details for the polynomial certificate

This appendix expands the interval argument behind theorem 5.2. For path Q, write $b_{Q,t} = \beta_{Q,t} - \beta_{P^\star,t}$. Its worst residual pairwise gap at radius r is

$$F_Q(r) = g_Q - r \sum_{b_{Q,t}<0} \frac{|b_{Q,t}|}{w_t} - \sum_{b_{Q,t}>0} b_{Q,t} \min\left\{ \frac{r}{w_t}, \widehat{d}_t - \ell_t \right\}. \tag{37}$$

The frozen radius is the first r for which $\min_{Q \in \mathcal{O}_k} F_Q(r) \le 0$.

Fix an interval with left endpoint δ_ℓ. If $b_{Q,t} > 0$ and $\delta_t \le \delta_\ell$, coordinate t has saturated at the floor throughout the interval and contributes $b_{Q,t}\ell_t$ to the affine path gap. If $b_{Q,t} > 0$ and $\delta_t > \delta_\ell$, it contributes $b_{Q,t}\widehat{d}_t - r b_{Q,t}/w_t$. If $b_{Q,t} < 0$, upward demand remains unsaturated and contributes $b_{Q,t}\widehat{d}_t + r b_{Q,t}/w_t$. These are exactly the arc terms in (15)–(16).

Let $\phi_\ell(r) = \min_{x \in X_k} F_x(r)$. If $\phi_\ell(\delta_\ell) \le 0$, the global first crossing is δ_ℓ because previous intervals had no crossing. Otherwise all positive-slope paths have a ratio strictly larger than δ_ℓ, and the smallest ratio solves (18). If it exceeds the right endpoint, no path crosses in the interval. At the last interval, every positive coefficient difference is floor-saturated; only negative differences can continue to reduce gaps as demand rises. If no denominator-positive path exists and the left endpoint is safe, the frozen radius is infinite.

The Charnes–Cooper feasible set is a scaled network-flow polytope. A feasible (y, τ) has $\tau > 0$ because $(b^\ell)^\top y = 1$; then $x = y/\tau$ satisfies $Nx = h$ and the objective equals the path ratio. Conversely every denominator-positive $x \in X_k$ maps to a feasible pair. Although the LP may return a convex combination at a multiple optimum, linear-fractional optimisation over a polytope has an optimal extreme point, hence an outside path.

B Exact disjunction as a union of polyhedra

For each outside path Q, define

$$\mathcal{D}_Q = \left\{ d \ge \ell : C_Q(d) \le C_P(d) \text{ for all } P \in \mathcal{I}_k \right\}.$$

Then the destabilising set is

$$\mathcal{D}_k = \bigcup_{Q \in \mathcal{O}_k} \mathcal{D}_Q.$$

Each $\mathcal{D}_Q$ is a polyhedron described either by exponentially many inside-path inequalities or by the compact extended potential formulation (20). The exact radius is the distance from $\widehat{d}$ to this union. This view clarifies three facts:

(i) the fixed-competitor problem is linear;
(ii) the difficulty lies in the union over outside paths and the moving best inside path; and
(iii) the frozen lower certificate replaces each $\mathcal{D}_Q$ by the larger half-space $\{d : C_Q(d) \le C_{P^\star}(d)\}$, so its union is larger and its distance is smaller.

C Independent checks encoded in the release

The test suite contains the following cross-checks.

1. Reduced arc costs match direct affine path coefficients; ordered slopes satisfy the Monge inequality.

2. The piecewise pairwise formula respects weighted norms and positive demand floors.
3. The polynomial lower method matches exhaustive outside-path enumeration on integer and continuous random instances.
4. Full-path lower certificates equal exact radii.
5. The compact fixed-competitor potential LP matches an independently coded LP with one inequality for every inside path.
6. The exact disjunctive radius and fixed-radius MILP agree on random small instances after a tolerance-separated boundary check.
7. The strict-gap example returns 1.5 for the lower certificate and 2 for the exact radius, and the continuous path relaxation fails.
8. Conformal order-statistic ranks and weighted revision scores match direct calculations.
9. Rolling realised regret never exceeds the deterministic theorem bound in the frozen and unit-test samples.

The script `scripts/verify_release.py` checks the frozen row counts, numerical invariants, analytical witness, figure files and example command-line output. It does not establish external independence because it was produced within the same workflow.

D Notation table

Symbol	Meaning
T	Number of periods.
K_i, p_i, h_i	Setup, unit production/purchase and one-period holding costs.
H_i, q_i	$H_i = \sum_{s<i} h_s$ and reduced slope $q_i = p_i - H_i$.
D_j	Cumulative demand before period j, $D_j = \sum_{t<j} d_t$.
$P^\star$	Selected nominal optimal path.
π_k	First k arcs of $P^\star$.
$\mathcal{I}_k, \mathcal{O}_k$	Paths inside and outside the protected prefix class.
$A_k(d), B_k(d)$	Best inside and outside reduced costs under revised demand d.
w, ℓ	Positive norm weights and coordinate-wise demand floor.
$\rho_{k,w}$	Exact strong prefix-stability radius.
$\underline{\rho}_{k,w}$	Polynomial frozen-comparator lower certificate.
κ_P, β_P	Constant setup cost and demand coefficient vector of path P.
g_Q, b_Q	Nominal path gap and coefficient difference relative to $P^\star$.
Γ_a	Exact weighted coefficient diameter of suffix paths from a.
$R_\pi(d)$	Bellman residual of an implemented block under realised demand.

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